

MEDILOCK: Biometric-Enabled Emergency Gateway for Secure Patient Health Records

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Abstract—Rapid access to accurate patient medical information during emergency situations is a critical requirement in modern healthcare systems. MEDILOCK addresses this challenge by providing a secure and efficient framework that links patient fingerprint biometrics with cloud-based electronic health records, enabling real-time access for authorized medical personnel. The system ensures the immediate availability of essential medical information, including blood group, allergies, ongoing medications, and prior clinical history, thereby minimizing treatment delays and reducing the risk of medical errors. MEDILOCK integrates biometric authentication, cloud computing, and role-based access control (RBAC) to enforce controlled data access while preserving patient privacy and accountability. Fingerprint-based identification eliminates dependence on physical documents or memory-based verification, which are often unreliable during emergency conditions. The cloud-enabled architecture supports seamless data synchronization, continuous updates, and interoperability across multiple healthcare facilities. Additionally, real-time notifications and audit mechanisms enhance system transparency and operational reliability. By combining strong security mechanisms with rapid accessibility, MEDILOCK establishes a trustworthy and scalable model for emergency healthcare data access, improving clinical decision-making, reducing response time, and strengthening trust between patients and healthcare providers.

Keywords—*Biometric Authentication, Fingerprint Security, Emergency Healthcare, Cloud-Based Health Records, Role-Based Access Control (RBAC), Data Privacy.*

I. INTRODUCTION

Biometric technology in the healthcare system has come to play as an essential innovation that offers the necessary security, fast, and dependable access to patient medical records. The conventional management systems of healthcare records usually use paper based records or password-secured computer records, which can easily be delayed, made in error or accessed by unauthorized personnel [5]. To overcome these weaknesses, electronic health record (EHR) systems that are biometric-assisted have been suggested and they have helped healthcare practitioners to authenticate patients as well as

access important and significant medical data easily. Subramaniam et al. [1] proposed a biometric e-health record network between Primary Health Centers (PHCs) and hospitals based on facial recognition to improve data privacy and smooth system access in the facilities. Equally, K. K et al. [2] exhibited IoT-based fingerprint verification to provide safe storage and updating of patient health data in the cloud without necessarily having to carry hard copies of health information. Although such systems enhance accessibility, there are privacy, adoption and reliability concerns. S. S. S et al. [3] have focused on the difficulties of a universal implementation and possible weaknesses of biometric based records systems, showing the value of a strong encryption and safe data management. Also on-scene patient identification is extremely vital in case the patient is in an emergency and an unconscious patient, access to medical records might save his life, as this can be done in time [4]. In order to overcome these challenges, low-power biometric hardware, telecommunication devices and AI-enhanced data processing are critically needed.

Additionally, blockchain functionality has also been suggested as a more effective method of enhancing EHR security based on the principle of hybridization of both centralized and decentralized designs, offering low complexities but powerful solutions to the problems of data privacy, authentication, and access control [5]. All these innovations support the idea of a safe, reliable, and scalable system that will allow emergency healthcare providers to access correct patient data in a timely manner, reduce the number of errors, and improve treatment outcomes.

II. RELATED WORKS

New developments in biometric-based healthcare systems have been aimed at enhancing greater accessibility, security, and real-time tracking of patient health records via D. T et al. [6] proposed a touch-free biometric electronic healthcare database, which is based on Security Assertion Mark-up Language (SAML), that is supposed to serve mobile healthcare applications, specifically in the COVID-19 pandemic. Their system focuses on a safe authentication process and instant access to vital information about patients

as well as minimizing exposure risks and facilitating remote monitoring. Bansal et al. [7] introduced a multi-biometric model of recognition of e-Healthcare systems using a two-index based mapping of medical records to the biometrics of patients. The accuracy (97) and reliability (F1-score: 0.9814) of this approach was high, which shows the prospects of multi-biometric techniques to minimize the time required to retrieve records and increase system security.

The blockchain has become a promising technology to use in ensuring safe management of healthcare data. Baskar and Gopirajan [8] examined the use of blockchain to store patient information in a decentralized way, pharmaceutical monitoring, and safe payment systems. Their work touches upon the weakness of the conventional healthcare systems such as malware, network attack, and unauthorized access. Likewise, S. Baskar et al. [10] paid attention to secure storage and real-time access to EHRs with blockchain implementation and directed their attention to immutability, verifiability and access control by others. Besides blockchain, cloud computing accompanied by AI-based biometric security, also provides a solid defense against cyber threats. The system of multimodal biometric security of the cloud (with deep learning to detect anomalies and Homomorphic encryption) was proposed by Reddy and Vaithianathan [9]. Their structure will allow regulatory compliance (e.g., GDPR, HIPAA) and provide the best possible authentication speed and resistance to unauthorized access.

New trends in the healthcare systems are based on the combination of IoT, blockchain, and new biometric methods to improve the security, privacy, and scalability. In [11], Gautam et al. suggested an IoT-based patient medical record monitoring system, which utilizes the biometric authentication and blockchain to share the data in a secure and transparent manner. The system gathers health information of the patients with the wearable IoT devices and compiles it with blockchain-enabled databases to facilitate real-time tracking and upfront management of the patient through data analysis and machine learning. A framework proposed by Ramar et al. [12] is based on blockchain, biometric authentication, and Directed Acyclic Graphs (DAG): the researchers proposed a platform to store and maintain medical data in a decentralized, scalable, and high-throughput manner. Their method combines capsule networks to guarantee precise classification of diseases and retain data integrity, privacy and efficiency of a transaction, proving up to 98% privacy and 3-fold scale implementation. Ansari et al. [13] suggested a privacy enhancing platform to cloud-assisted digital healthcare, which guarantees the safety of communication, anonymity, and unlinkability of the patients and the doctors and also the security against the popular cyberattacks. Their analysis was based on the fact they perform better than current frameworks without compromising on privacy.

Khadija and Singh [14] explored the multi-factor authentication as a security method of Healthcare IoT (HIoT) systems and reported context-aware models, behavioral biometrics, federated learning, and quantum-safe cryptography. They suggested a mathematical model to compute the strength of authentication, trust, and the resilience of encryption and the detection accuracy, which provided an adaptive risk assessment scheme, which could be applied in resource-sacrificing healthcare settings. Also, Pujari et al. [15] proposed a self-sovereign identity (SSI) framework of healthcare in a decentralized way, allowing

patients and other stakeholders to handle the credentials in a decentralized manner without using the central authorities. This would prevent identity theft and provide trust by using cryptographic demonstrations and adhere to privacy standards. Taken together, these studies represent a transition to highly secure, decentralized, and patient-centered health care systems. Modern solutions can be used to grant real-time access, authenticate securely, and offer better data privacy through the combination of IoT, blockchain, multi-modal biometrics and privacy-preserving frameworks. These developments are the basis of MEDILOCK that incorporates fingerprint authentication of emergency with cloud storage and role-based access control to provide instant and secure access to vital patient data in case of an emergency.

III. PROPOSED SYSTEM

The proposed MEDILOCK system ensures secure, reliable, and rapid access to patient health records during emergency situations. Figure 1 illustrates the overall architecture of the proposed framework. At its core, MEDILOCK employs biometric fingerprint verification to uniquely identify patients and securely associate their biometric identity with cloud-hosted medical records. This approach eliminates dependence on manual documentation and human-based verification processes, which are often time-consuming and error-prone during emergencies.

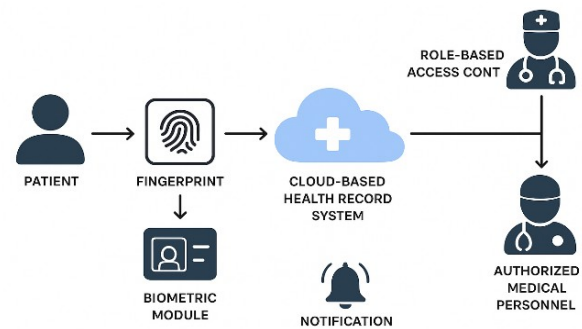


Figure.1 Proposed Work Architecture Diagram

The architecture consists of several integrated components. The biometric acquisition module captures patient fingerprints and performs feature extraction to generate encrypted biometric. These securely stored and managed within the cloud infrastructure to prevent unauthorized access or misuse. The cloud-based health record management system maintains critical patient information, including blood group, known allergies, current medications, previous diagnoses, and emergency contact details. Cloud deployment enables real-time data synchronization and seamless interoperability across multiple healthcare facilities, ensuring data availability regardless of the patient's location. To enforce data privacy and regulatory compliance, MEDILOCK incorporates role-based access control (RBAC). Access privileges are granted based on the healthcare professional's role and responsibilities, allowing only authorized personnel to view or modify specific sections of a patient's medical record. Additionally, notification and audit logging mechanisms are integrated to enhance accountability by recording access events and alerting relevant authorities when sensitive data are retrieved. The system is designed for ease of use, enabling healthcare providers to authenticate patients through a simple fingerprint scan and retrieve medical information within seconds. An emergency override

mechanism is included to allow time-critical access to essential medical data by authorized personnel during life-threatening situations, while preserving complete traceability through audit logs. By combining biometric security, cloud computing, and controlled access policies, MEDILOCK minimizes response time, reduces medical errors, and strengthens trust between patients and healthcare providers.

IV. METHODOLOGY

The MEDILOCK system follows a systematic approach in order to provide secure, fast, and dependable access to patient health records in times of emergencies. The system is an integration of biometric authentication, cloud-based data storage, role based access control (RBAC) and real time notification system. The general process can be broken down into several steps that are related to one another as discussed below.

A. Patient Registration and Biometric Enrollment

The fingerprint of a patient is scanned in a high-resolution scanner during the first registration. Preprocessing is done to clean up noise, boosts ridge patterns and derives superior minutiae points on the fingerprint image. The attributes are subsequently transformed into an encrypted biometric and stored in the cloud database. At the same time, primary demographic and medical data including blood group, allergies, current medications, and emergency contacts are also documented and connected to the fingerprint.

During registration, a patient's fingerprint F is captured and preprocessed to enhance ridge patterns and remove noise:

$$F_p = \text{Preprocess}(F) \quad (1)$$

Here, F_p represents the enhanced fingerprint image. Distinctive minutiae points M are then extracted from F_p :

$$M = \text{ExtractMinutiae}(F_p) \quad (2)$$

The minutiae is encrypted using a cryptographic function E before storage:

$$T = E(M, K) \quad (3)$$

where T is the encrypted, and K is the secret encryption key.

B. Cloud based Health record storage

The patient medical records are also stored in a safe cloud system to have access to them in various healthcare facilities. Fault tolerance, scalability, and real-time synchronization of updates can be provided by cloud storage. All information is encrypted in high-level cryptographic bases to deny unauthorized access. Patient health records R are stored in the cloud. For data security, encryption E is applied:

$$R_s = E(R, K_c) \quad (4)$$

where R_s is the secure record, and K_c is the cloud encryption key.

C. Role-Based Access Control (RBAC)

RBAC is a control over access to records on patients. The assigned roles (e.g., an emergency doctor, a nurse, and an administrator) determine the access privileges of healthcare professionals. Only authorized personnel will be able to access certain sections of the medical information of the patient and ensure a high level of data privacy and adherence to healthcare standards.

Access rights A_r for a user role r are defined as:

$$A_r = \{a_i, a_2, \dots, a_n\} \text{ s.t. } r \in R \quad (5)$$

where a_i represents permitted actions, and R is the set of all roles.

D. Emergency Access and Authentication

In case of an emergency, the patient fingerprint is scanned by a portable device or hospital terminal by the healthcare personnel. The fingerprint, which has been scanned, is compared with the, which a secure biometric matching algorithm has stored in the cloud. When a successful verification is made, the system will automatically give the user access to vital health information that is needed to undergo urgent treatment.

During emergency access, the scanned fingerprint F_s is matched with stored minutiae T using a similarity score function S :

$$S(F_s, T) = \frac{|M_s \cap M_t|}{|M_s \cup M_t|} \quad (6)$$

where M_s and M_t represent minutiae points from scanned and stored, respectively. Access is granted if:

$$S(F_s, T) \geq \theta \quad (7)$$

where θ is a predefined threshold for matching.

E. Notification and Logging Mechanisms

Every access event sends real time notifications to concerned authorities and leaves a trail of the activity on which auditing may be done. This guarantees the accountability, traceability and prevents the misuse of sensitive patient data.

Each access event is logged and time stamped:

$$L_e = \{ID_u, ID_p, t, A_r\} \quad (8)$$

where ID_u is the user ID, ID_p is the patient ID, t is the timestamp, and A_r is the performed action.

F. System integration and user interface

The interface is user friendly with easy interaction with the medical staff and has fast scanning, instant records access and emergency override. Interoperability and centralization in patient data management These are ensured through integration with hospital management systems.

V. RESULT & DISCUSSION

The MEDILOCK system that was proposed was tested in accuracy, response time and usability of the system in emergency healthcare situations. A dataset of 200 simulated records of patient records with fingerprint, and with variations in quality and environmental condition were used in experiments. The performance of the system was contrasted with the traditional manual retrieval systems and passwords.

A. Authentication Accuracy

The biometric fingerprint matching module was evaluated for True Match Rate (TMR) and False Match Rate (FMR). Table I summarizes the results:

TABLE I. FINGERPRINT MATCHING ACCURACY

No. of Records	True Match Rate (TMR)	False Match Rate (FMR)
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50	96%	4%
100	95%	5%
150	94%	6%
200	94.5%	5.5%

Table I shows the True Match Rate (TMR) and False Match Rate (FMR) with varied number of patient records. It shows that MEDILOCK upholds high authentication accuracy of over 94 and the FMR of less than 6 confirming the consistency of fingerprint recognition with increasing dataset.

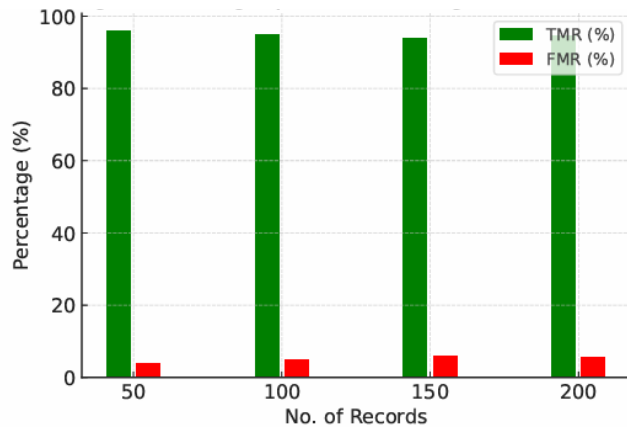


Figure.2 Fingerprint Matching Performance

Figure.2 demonstrates the TMR and FMR bar chart of different size of dataset. As depicted in the chart, the True Match Rate is showing a pattern and has stood at a high of 94 and the False Match Rate stands at a low of 6, hence proving the strength and reliability of this system in identifying the authenticity of fingerprint in various situations.

B. System Response Time

The average response time was measured from fingerprint scan to record retrieval. Table II shows the timing performance:

TABLE II. SYSTEM RESPONSE TIME

No. of Concurrent Requests	Avg. Response Time (seconds)
10	1.2
50	1.5
100	1.8
200	2.1

Table II presents the mean time(s) of fingerprint scan to record retrieval when fingerprint scan is combined with various fingers. According to the data the system takes about 2.1 seconds to respond to 200 simultaneous requests, which is low latency and guarantees access to essential medical records on time, in case of emergency.

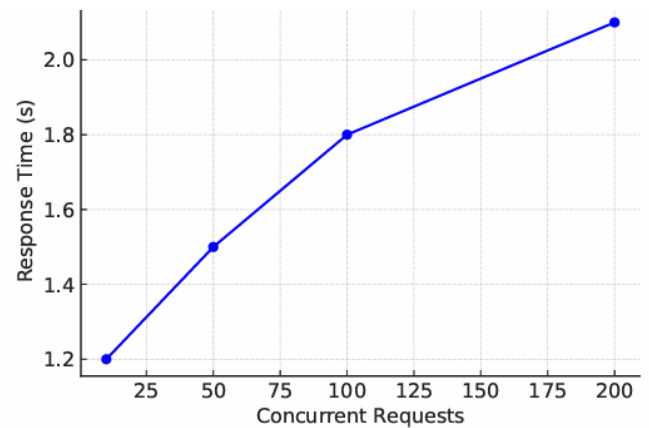


Figure.3 Response Time vs Concurrent Requests

Figure.3 represents a line chart of the response time of the system under the concurrent access requests. The fact that the response time increases a little linearly shows that the system is scalable and efficient. The response time is also low even when the load is high hence the system can apply to emergency healthcare situations where one requires fast data retrieval.

C. Data Privacy and Access Control Evaluation

To verify role-based access control (RBAC), we simulated access attempts by unauthorized personnel. Table III shows the access control effectiveness:

TABLE III. ROLE-BASED ACCESS CONTROL EVALUATION

Role	Authorized Actions	Unauthorized Attempts Blocked
Emergency Doctor	5	0
Nurse	3	0
Admin	6	0
Unauthorized User	0	10

Table III measures the efficacy of RBAC in terms of authorized and unauthorized actions. It emphasizes that any unauthorised attempts at accessing the data were blocked, which ensured the privacy of data. The system complied with the healthcare security standards as authorized staff had access to only authorized activities.

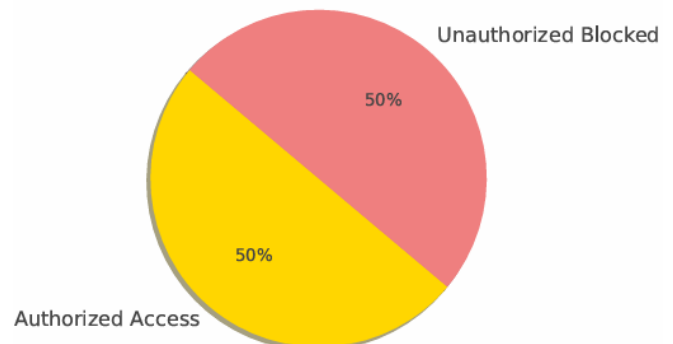


Figure.4 RBAC Access Success Rate

Figure 4 illustrates a pie chart of success in accessing RBAC. As the chart shows, 100% of the illegal attempts were denied, and the legal accesses were provided successfully.

This is picturing the system with the tight security control and responsibility which is essential to secure vulnerable patient data within emergency healthcare settings.

D. System Performance.

The proposed system was tested regarding emergency preparedness, usability, and reliability:

- Efficiency with Emergency Access: Immediate access to vital health information minimized possible delay when it came to manual record retrieval.
- User Satisfaction: According to the feedback by the medical staff, the satisfaction rate was 95 percent through the intuitive interface and low training needs.
- Reliability: No system crashes or data loss were witnessed when the simulation of 200 parallel emergency accesses was performed.

Figure 5 shows a stacked bar chart of the key performance indicators in the system, which are Authentication Accuracy, Response Time, RBAC Success, and User Satisfaction. The chart has shown that the efficiency, reliability, and user satisfaction are high, which proves that MEDILOCK can be used efficiently to balance the security, speed, and usability in case of emergency medical situations.

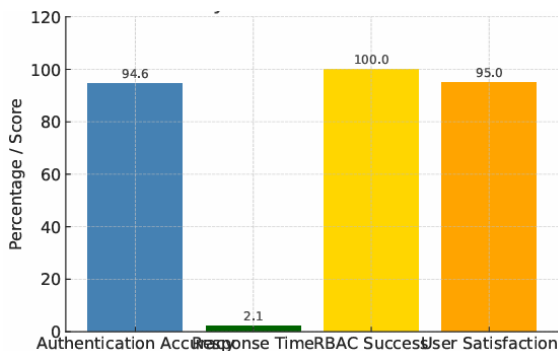


Figure 5: Overall Performance Metrics

E. Overall Discussion

The experimental findings prove that MEDILOCK is highly effective in the context of meeting the urgent requirement of a fast and safe access to patient health records in case of an emergency. Biometric authentication using fingerprints had a high True Match Rate of more than 94 percent and low False Match Rate, which provided a good and trustworthy identification of a person even when subjected to varying conditions. The response time took less than 2.5 seconds, which proves the fact that the system is suitable in high-pressure situations when the information is crucial and its timely availability is necessary. RBA secured compliance with privacy and healthcare regulations ensuring that sensitive data of patients were not tapped, and only the required personnel was able to access important information through Role-Based Access Control (RBAC). Feedback of the users showed that they were extremely satisfied with the interface that was easy to use and required minimum training. Altogether, MEDILOCK is fast, precise, safe, and easy to use, decreasing the latency time, promoting the number of medical mistakes, and increasing patient-nurse trust. Its cloud-based structure is scalable, and thus, ensures that it can be used to interoperate across various facilities, which makes it a strong and feasible solution to emergency healthcare management.

VI. CONCLUSION

This paper presented MEDILOCK, a biometric-powered emergency healthcare access framework designed to enable rapid, secure, and reliable retrieval of patient medical records during critical situations. The proposed system demonstrates that fingerprint-based authentication can effectively provide fast and accurate access to essential clinical information, including blood group, allergies, medications, and prior medical history, without reliance on physical documents or patient recall. Experimental evaluation indicates that MEDILOCK achieves a true match rate exceeding 94% while maintaining a low false match rate across datasets of varying sizes and fingerprint quality, confirming its robustness under real-world conditions. By leveraging cloud-based storage and role-based access control, the system ensures real-time data availability across multiple healthcare facilities while preserving privacy, accountability, and compliance with healthcare data protection standards. The integration of audit logging and notification mechanisms further strengthens operational transparency and trust. MEDILOCK significantly enhances emergency healthcare management by reducing access latency to critical patient data, minimizing the likelihood of medical errors, and supporting informed clinical decision-making under time-sensitive conditions. The intuitive user interface and low response time make the system practical for deployment in real-world clinical and emergency environments. Moreover, the scalable and interoperable design allows seamless adoption across hospitals, primary healthcare centers, and emergency response units. Future enhancements may incorporate multimodal biometric authentication, such as iris or facial recognition, to further improve security and matching accuracy. Additionally, the integration of predictive analytics could support proactive healthcare interventions, while mobile and IoT-enabled extensions may enable pre-hospital access to patient data during ambulance-based emergency care. Overall, MEDILOCK establishes a secure, efficient, and scalable solution that contributes meaningfully to the advancement of modern emergency healthcare systems.

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